Further Calculus

Questions 1–57

1.14 — Question 1. Find the mean value of each of the following on the given intervals:

a) $f(x) = 5x - 2$ on the interval $[3, 8]$

b) $f(x) = 5x^2$ on the interval $[2, 5]$

c) $f(x) = \frac{1}{\sqrt{x}}$ on the interval $[9, 16]$

1.14 — Question 2. Find the mean value of the function

$$f(x) = 2x^2 - 1$$

in the interval $-0.5 \leq x \leq 1.2$, giving your answer to 3 significant figures.

1.14 — Question 3. Show that the mean value of

$$f(x) = x^2$$

on the interval $[0, k]$ where $k > 0$ is $\frac{1}{3}k^2$

1.14 — Question 4. Find the mean value of the function

$$f(x) = 8x^3$$

in the interval $0 \leq x \leq a$, giving your answer in terms of a

1.14 — Question 5. Show that the mean value of

$$f(x) = mx + c$$

on the interval $[a, b]$ where m, c, a and b are constants and $a < b$ is

$$\frac{1}{2}m(a + b) + c$$

1.14 — Question 6. Find the mean value of the function

$$f(x) = (x + 4)^2$$

in the interval $a \leq x \leq 2a$, giving your answer in the form $pa^2 + qa + r$ where p, q and r are constants to be found.

1.14 — Question 7. Find the mean value of the function

$$f(x) = \frac{1+x}{\sqrt{x}}$$

in the interval $0 \leq x \leq k$, giving your answer in the form

$$\frac{p}{\sqrt{k}} + \frac{p\sqrt{k}}{q}$$

where p and q are constants to be found.

1.14 — Question 8. The mean value of the function

$$f(x) = 2\sqrt{x} + k$$

in the interval $[1, 9]$ where k is a positive constant is $\frac{22}{3}$
Find the value of k

1.14 — Question 9. The mean value of the function

$$f(x) = x^2 + 1$$

in the interval $[2, 5]$ is half the mean value of $f(x)$ for the interval $[0, k]$ where k is a positive integer. Find the exact value of k

1.14 — Question 10. The mean value of the function

$$f(x) = 3x^2 + 12x - 40$$

in the interval $[0, k]$ where k is a positive constant is 51
Find the exact value of k

1.14 — Question 11. The mean value of the function

$$f(x) = ax - 4$$

is -5.5 over the interval $[-2, 1]$

a) Determine the value of a

It is known that $f(x)$ also has a mean value of 18.5 over the interval $[p, 2p]$

b) Determine the value of p

1.14 — Question 12. The velocity of a particle after t seconds is given by

$$v = \frac{1}{6}\sqrt{t}\left(12 - \frac{1}{t}\right)$$

a) Show that the mean velocity for $1 \leq t \leq 3$ is 2.68 ms^{-1} to 3 significant figures.

b) Calculate the mean acceleration of the particle for the same time period giving your answer to 3 significant figures.

1.14 — Question 13. The temperature of a metal rod within some machinery at a distance x metres from one end is modelled by $\mathrm{T}(x) = -5x^3 + 12x^2 + 4x$ where $\mathrm{T}(x)$ is the temperature in degrees Celsius and x is the distance in metres from one end of the rod. The rod is 3 metres long. Find the mean temperature of the rod over its entire length.

1.14 — Question 14. The growth of a population of rabbits $\mathrm{P}(t)$ during the first 12 months of the year is modelled by $\mathrm{P}(t) = 2t^3 - 7t^2 + 15t + 40$ where $\mathrm{P}(t)$ represents the number of rabbits and t is the number of months that have passed. Find the average population of rabbits over the first 12 months.

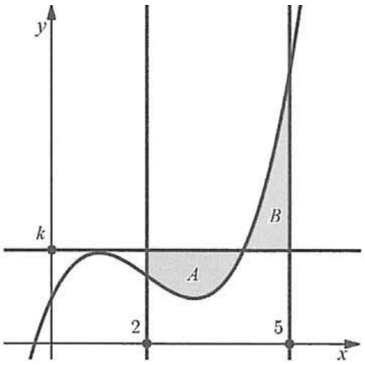
1.14 — Question 15.

The diagram below shows the curve

$$f(x) = x^3 - 6x^2 + 9x + 4$$

with shaded regions A and B enclosed by the lines

$$y = k, x = 2 \text{ and } x = 5$$



Given that the areas A and B are equal, find the value of k

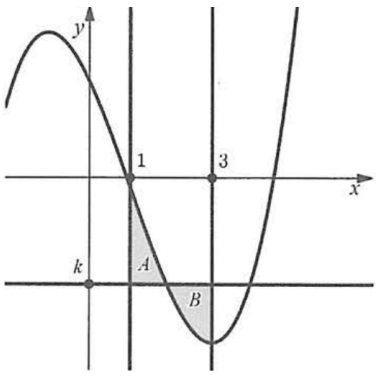
1.14 — Question 16.

The diagram below shows the curve

$$f(x) = x^3 - 3x^2 - 9x + 10$$

with shaded regions A and B enclosed by the lines

$$y = k, x = 1 \text{ and } x = 3$$



Given that the areas A and B are equal, find the value of k

1.14 — Question 17. A function

$$f(x) = x^3 - 2$$

has a mean value of

$$\frac{195}{4}$$

over the interval $[2, k]$ where k is a positive constant.
Find the exact value of k

1.14 — Question 18. Given that the function f is defined by

$$f(x) = x^2 + x - 1 \text{ for } 2 \leq x \leq 4$$

a) Find the mean value of f

Another function g is defined by

$$g(x) = f(x) + c \text{ for } 2 \leq x \leq 4$$

b) Given that the mean value of g is 25, find the value of c

1.14 — Question 19. A function f has mean value k over the interval

$$0 \leq x \leq 4$$

where k is an unknown constant.
In each case, find the mean value over the given interval in terms of k

a) $g(x)$ which is a translation of $y = f(x)$ by $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ over the interval $-2 \leq x \leq 2$
b) $h(x)$ which is a reflection in the x-axis of $y = f(x)$ over the interval $0 \leq x \leq 4$
c) $m(x)$ which is a reflection in the y-axis of $y = f(x)$ over the interval $-4 \leq x \leq 0$
d) $n(x)$ which is a translation of $y = m(x)$ by $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ over the interval $-3 \leq x \leq 1$

1.14 — Question 20. A function f has mean value k over the interval

$$2 \leq x \leq 7$$

where k is an unknown constant.
In each case, find the mean value over the given interval in terms of k

a) $g(x)$ which is a translation of $y = f(x)$ by $\begin{pmatrix} -1 \\ 4 \end{pmatrix}$ over the interval $1 \leq x \leq 6$
b) $h(x)$ which is a stretch, parallel to the x-axis, scale factor 3 of $y = f(x)$ over the interval $6 \leq x \leq 21$
c) $m(x)$ which is a stretch, parallel to the y-axis, scale factor $\frac{1}{2}$ of $y = f(x)$ over the interval $2 \leq x \leq 7$

1.14 — Question 21. The root mean square of a function $f(x)$ defined for $a \leq x \leq b$ over this interval is

$$\sqrt{\frac{1}{b-a}\int_a^b [f(x)]^2 \, \mathrm{d}x}$$

Find the root mean square of

$$f(x) = -x(x-0.5)(x-1)$$

from $x = 0.1$ to $x = 0.6$ giving your answer to 3 significant figures.

1.15 — Question 1. Find y^2 for each of the following:

a) $y = x + 1$
b) $y = 3x - 1$
c) $y = x - \frac{1}{x}$
d) $y = x^2 + 5$
e) $y = 2\sqrt{x} + 3$
f) $y = \frac{1}{\sqrt{x^3}} + 1$

1.15 — Question 2. Find the exact volume created by rotating the curve through 2π radians around the x-axis for each of the following:

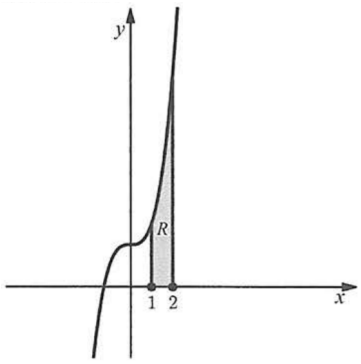
a) $y = x - 1$ for $1 \leq x \leq 5$
b) $y = x^2 + 2$ for $0 \leq x \leq 2$
c) $y = x^{-\frac{3}{2}}$ for $1 \leq x \leq 3$
d) $y = 2 + \sqrt{x}$ for $1 \leq x \leq 4$

1.15 — Question 3.

The side of a solid may be modelled by the curve

$$y = x^3 + 2$$

for $1 \leq x \leq 2$ where x and y are measured in centimetres and the x-axis is a line of symmetry when the area is rotated 2π radians about the x-axis.



Find the volume of the solid giving your answer to 1 decimal place.

1.15 — Question 4. A region, R, is bounded by the curve

$$y = 4 - x^2$$

and the x-axis. The region is rotated through 2π radians about the x-axis to form a solid of revolution.

a) Sketch the region R

b) Find the exact volume of the solid formed.

1.15 — Question 5.

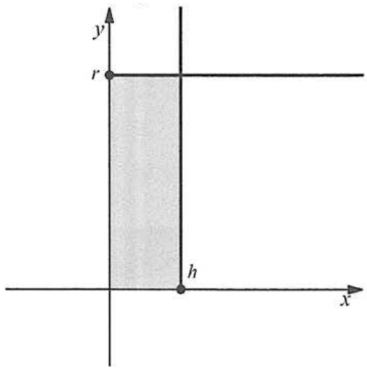
The diagram below shows the region enclosed by part of the horizontal line

$$y = r$$

and part of the vertical line

$$x = h$$

where h and r are positive constants.



The line $y = r$ is rotated 2π radians about the x-axis. Show that the volume of the solid generated is

$$\pi r^2 h$$

1.15 — Question 6.

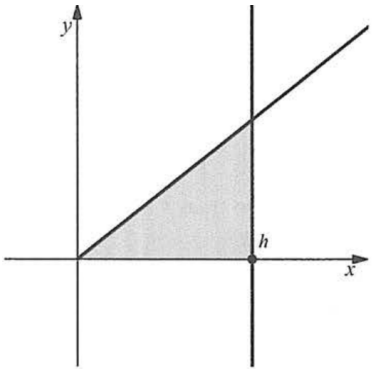
The diagram below shows the region enclosed by part of the line

$$y = kx$$

and the vertical line

$$x = h$$

for positive constants h and k. The line $y = kx$ is rotated 2π radians about the x-axis, forming a cone with a radius of base r

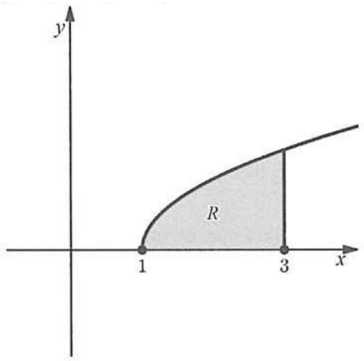


a) Find k in terms of r and h
b) Hence show that the volume of the solid generated is

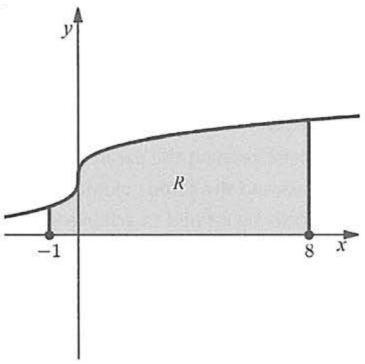
$$\frac{1}{3}\pi r^2 h$$

1.15 — Question 7.

For each of the following, the shaded region shown in the diagram is bounded by the given lines, the curve and the x-axis.
In each case, find the exact volume of the solid generated when the region is rotated 2π radians about the x-axis:
a) $y = \sqrt{x-1}$, $x = 1$, $x = 3$



b) $y = 2 + x^{\frac{1}{3}}$, $x = -1$, $x = 8$



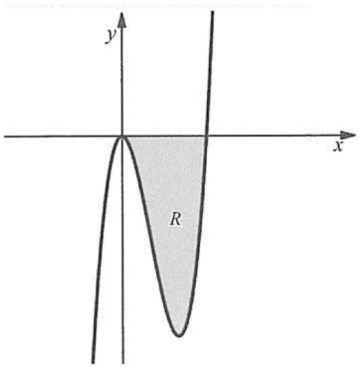
c) $y = -2x^2 + 6x$ and the x-axis.

1.15 — Question 8.

The shaded region below is bounded by the curve

$$y = x^3 - 4x^2$$

and where the curve meets the x-axis.



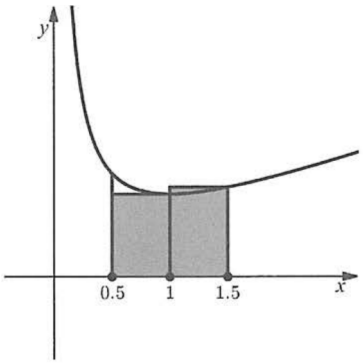
The region is rotated around the x-axis until a solid of revolution is generated. Find the volume of the solid generated giving your answer to 1 decimal place.

1.15 — Question 9.

The curve

$$y = x + \frac{1}{x}$$

for $x > 0$ is shown in the graph below.



Alfie wants to find the volume created by rotating the shaded region 2π radians about the x-axis. He decides to approximate the volume by finding the sum of the volume of two individual cylinders.

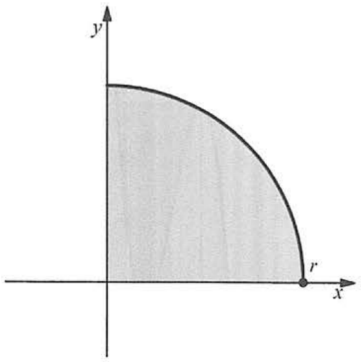
a) Find the radius of the two cylinders he finds by choosing $x = 1$ and $x = 1.5$, and hence find the approximate volume.
b) Explain how, without using Calculus, Alfie could improve the approximation.
c) Use Calculus to find the exact volume.

1.15 — Question 10.

The diagram below shows part of an equation of a circle given by

$$x^2 + y^2 = r^2$$

The region is restricted such that $x \geq 0$ and $y \geq 0$



The region is rotated around the x-axis until a solid of revolution is generated.

a) Find the volume of the solid generated.

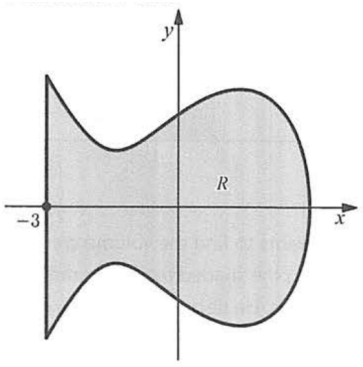
b) Hence show that the volume of a sphere is given by $V = \frac{4}{3}\pi r^3$

1.15 — Question 11.

The diagram below shows the equation of a curve given by

$$2y^2 - 6x = 9 - x^3$$

and bound by the line $x = -3$ and where the curve meets the x-axis.



a) Find the value at which the curve meets the x-axis.

The region is rotated around the x-axis until a solid of revolution is generated.

b) Find the volume of the solid generated.

Prerequisite knowledge for Q12-Q13: Conics

1.15 — Question 12. A bowl is modelled using the curve

$$y^2 = 3x$$

and rotating the curve about the x-axis.

a) Sketch the graph of $y^2 = 3x$

The bowl needs to have a depth d such that it can hold exactly 500 cm^3 of water.

b) Determine the depth of the bowl to the nearest millimetre.

c) What assumptions have been made in using this model?

1.15 — Question 13. A wine glass may be modelled using the curve

$$y^2 = kx$$

and rotating the curve about the x-axis, where k is a positive constant. The wine glass is to have a volume of 180 cm^3 and a height of 6 cm.

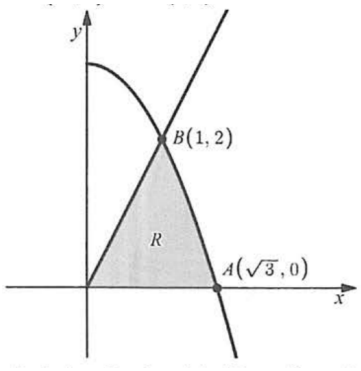
a) Determine the maximal radius of the wine glass to the nearest millimetre.

It is decided the diameter of the top of the wine glass is too big.

b) Whilst keeping the parabola shape of the glass and the same volume, how could the model be refined to achieve this?

1.15 — Question 14.

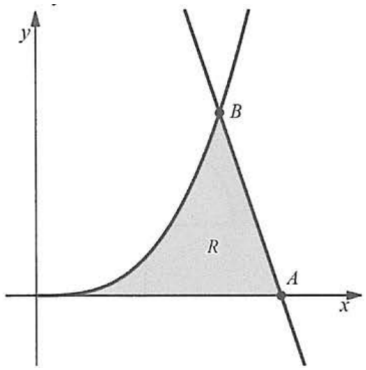
The diagram below shows the region enclosed by part of the line $y = 2x$ and the curve $y = -x^2 + 3$ and the x-axis. They meet at points $A(\sqrt{3}, 0)$ and $B(1, 2)$



The shaded region is rotated 2π radians about the x-axis. Find the exact volume of the solid of revolution formed.

1.15 — Question 15.

The diagram below shows the region enclosed by part of the curve $y = \frac{1}{4}x^3$ and the straight line $3x + y = 8$ and the x-axis.



The point A is where $3x + y = 8$ meets the x-axis. The point B is where the curves $y = \frac{1}{4}x^3$ and $3x + y = 8$ meet.

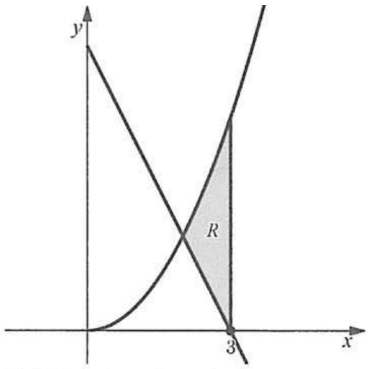
a) Find the coordinates of points A and B

The shaded region is rotated 2π radians about the x-axis.

b) Find the exact volume of the solid of revolution formed.

1.15 — Question 16.

The diagram below shows the region enclosed by part of the curve $y = 0.5x^2$, the straight line $2x + y = 6$ and the line $x = 3$



a) Find the point where the curve $y = 0.5x^2$ and $2x + y = 6$ meet.

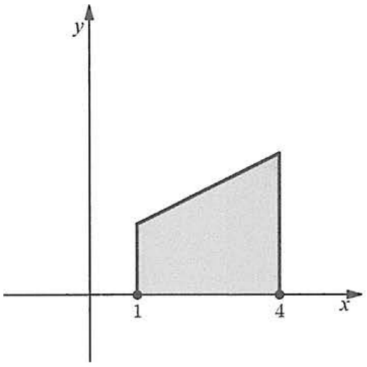
The shaded region is rotated 2π radians about the x-axis.

b) Find the exact volume of the solid of revolution formed.

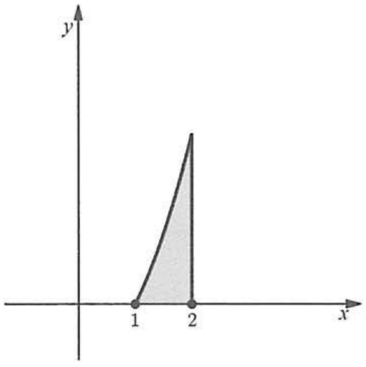
1.15 — Question 17.

Find the exact volume created by rotating the curve around the x-axis by the given angle for each of the following:

a) $y = \frac{1}{2}x + 1$ for $1 \leq x \leq 4$ by $90°$



b) $y = x^2 - 1$ for $1 \leq x \leq 2$ by π radians

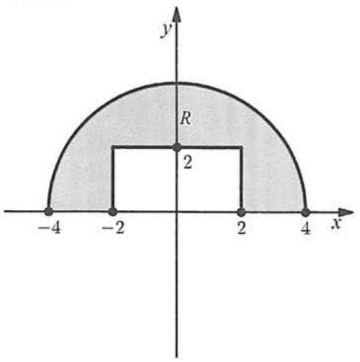


c) $y = 4 - \sqrt{x}$ for $0 \leq x \leq 1$ by $\frac{\pi}{2}$ radians



1.15 — Question 18.

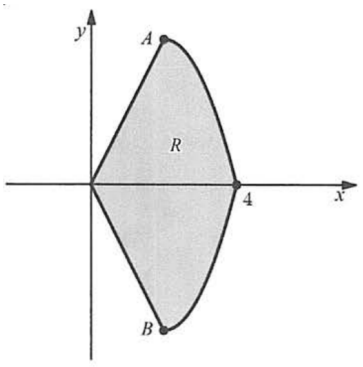
The diagram shows the region enclosed by the curve $x^2 + y^2 = 16$, the x-axis, the lines $x = -2$ and $x = 2$ for $0 \leq y \leq 2$ and the line $y = 2$ for $-2 \leq x \leq 2$



The shaded region is rotated about the x-axis to find a solid of revolution.
Using Calculus, determine the exact volume of the solid of revolution formed.

1.15 — Question 19.

The diagram below shows the region enclosed by parts of the curves $y = 2x$, $y = -x^2 + 4x$, $y = x^2 - 4x$ and $y = -2x$
The curves $y = 2x$ and $y = -x^2 + 4x$ meet at the point A
The curves $y = x^2 - 4x$ and $y = -2x$ meet at the point B

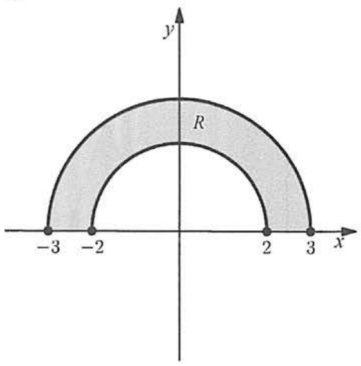


a) Find the coordinates of point A
b) Find the coordinates of point B

The shaded region is rotated about the x-axis to find a solid of revolution.
c) Find the exact volume of the solid of revolution formed.

1.15 — Question 20.

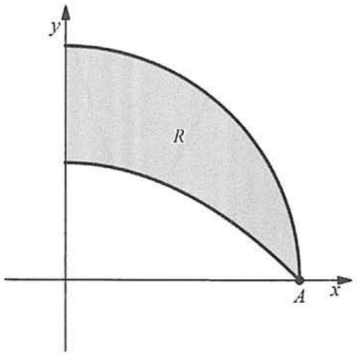
The diagram below shows the region enclosed by the curve $x^2 + y^2 = 9$, $x^2 + y^2 = 4$ and the x-axis.



The shaded region is rotated about the x-axis to find a solid of revolution.
Determine using Calculus, the exact volume of the solid of revolution formed.

1.15 — Question 21.

The diagram below shows the region enclosed by the curve $y = \sqrt{16 - x^2}$, $y = -\frac{1}{8}x^2 + 2$ and the y-axis.
The curves $y = \sqrt{16 - x^2}$ and $y = -\frac{1}{8}x^2 + 2$ meet at the point A
Point A lies on the x-axis.



a) Find the coordinates of the point A

The shaded region is rotated about the x-axis to find a solid of revolution.

b) Find the exact volume of the solid of revolution formed.

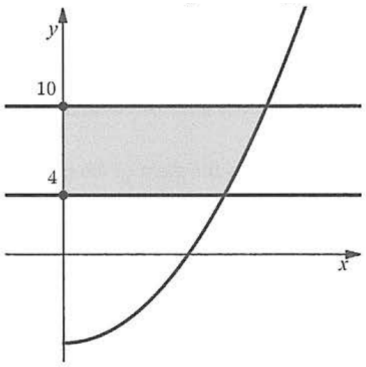
Rotations about the y-axis

1.15 — Question 22. Find the exact volume created by rotating the curve through 2π radians around the y-axis for each of the following:

a) $x = y + 2$ for $1 \leq y \leq 5$
b) $x = y^3$ for $0 \leq y \leq 1$
c) $y = x^2$ for $0 \leq y \leq 4$

1.15 — Question 23.

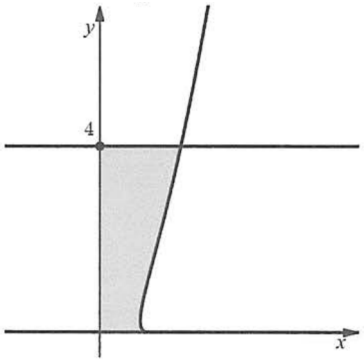
The diagram below shows the region enclosed by part of the curve $y = x^2 - 6$ where $x \geq 0$ and the horizontal lines $y = 4$ and $y = 10$



The shaded region is rotated 2π radians about the y-axis. Show that the volume of the solid generated is 78π units^3.

1.15 — Question 24.

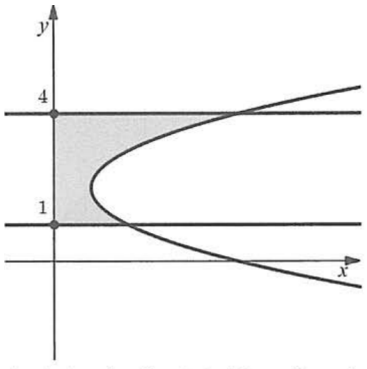
The diagram below shows the region enclosed by part of the curve $x^2 = y - \sqrt{y} + 1$, the x-axis and the line $y = 4$



The shaded region is rotated 2π radians about the y-axis. Find the exact volume of the solid generated.

1.15 — Question 25.

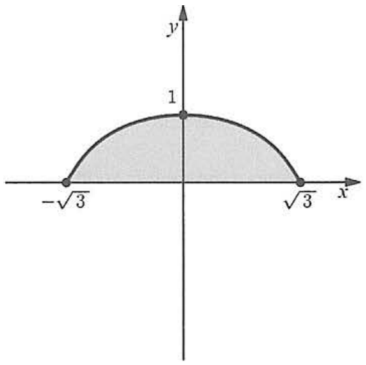
The diagram shows the region enclosed by $x = y^2 - 4y + 5$ and the horizontal lines $y = 1$ and $y = 4$



The shaded region is rotated 2π radians about the y-axis.
Find the exact volume of the solid generated.

1.15 — Question 26.

The diagram below shows the equation of a curve given by $y^3 + 2y = 3 - x^2$ and where the curve meets the x-axis.



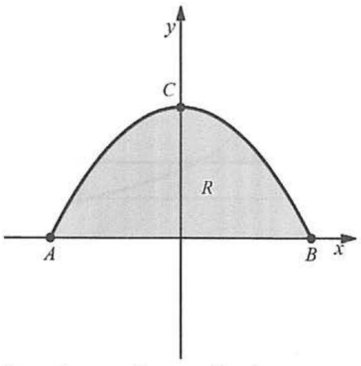
The region is rotated around the y-axis until a solid of revolution is generated.
Find the exact volume of the solid generated.

1.15 — Question 27. Show that when rotated by 2π radians about the y-axis, the line $x = r$ for $0 \leq y \leq h$ where r and h are positive constants creates a volume of $\pi r^2 h$ units^3.

1.15 — Question 28. The line $y = kx$ for a positive constant k, where $0 \leq y \leq h$, is rotated by 2π radians about the y-axis, forming a cone with a radius of base r. Show that the volume of the solid is $\frac{1}{3}\pi r^2 h$ units^3.

1.15 — Question 29.

A large pile of sand is formed at a construction site. The diagram below shows a model for the vertical cross-section of the sand pile. The shaded region is bounded by the curve $y = 1 - x^2$ and the x-axis.



a) State the coordinates of points A, B and C

To model the volume of the sand pile, the shaded region is rotated about the y-axis to form a solid of revolution. It is given that 1 unit is equivalent to 1 metre.

b) Use the model to estimate the volume of sand in the pile. Give your answer to 3 significant figures.

The density of the sand is 1600 kg/m^3.

c) Estimate the mass of the sand pile giving your answer to 3 significant figures.

d) Comment on the validity of your answer to part b)

1.15 — Question 30. A wine glass whose surface outline may be modelled using the function

$$y = 0.4x^2$$

is rotated about the y-axis to form its bowl.

a) Show, that the volume of liquid V cm^3 with depth d cm is given by $V \approx 3.93d^2$ cm^3.

The wine glass is filled to a depth of 6 cm.

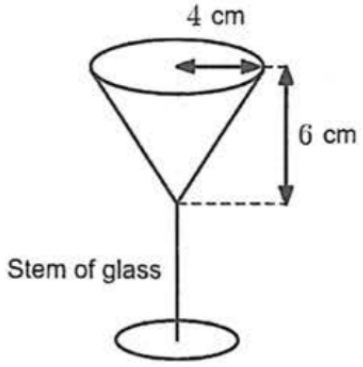
b) Estimate the volume of wine in the glass, giving your answer to 3 significant figures.

The wine glass holds 360 cm^3 of liquid.

c) Estimate the maximum depth of the wine glass bowl, giving your answer to 3 significant figures.

1.15 — Question 31.

A conical cocktail glass has a maximum cross-sectional radius of 4 cm and depth of 6 cm. The stem of the glass is 8 cm.



Show, without using Calculus,

a) that the maximum volume of liquid V cm^3 it can hold is approximately 101 cm^3.

b) the glass will be half full when the depth of the liquid is approximately 4.76 cm.

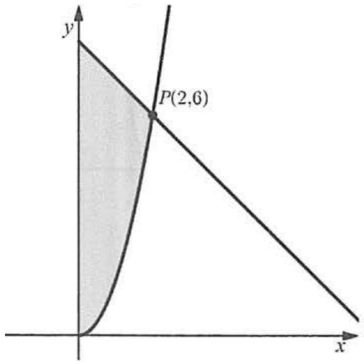
By considering the stem of the glass as lying on the y-axis, with the origin being where the stem connects with the base of the glass,

c) Find a linear equation to model the radius of the glass, x cm, in terms of y cm, the distance from the x-axis.

d) Hence, show using Calculus that the maximum volume of liquid V cm^3 that it can hold is approximately 101 cm^3.

1.15 — Question 32.

The diagram below shows the region enclosed by part of the curve $y = 1.5x^2$, the straight line $x + y = 8$ and the y-axis.



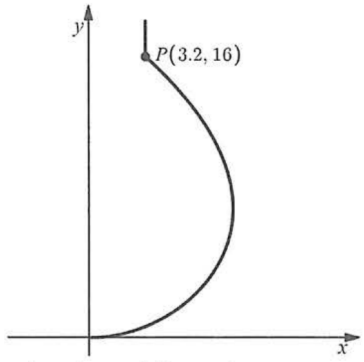
The curve and the straight line meet at $P(2, 6)$.
The shaded region is rotated 2π radians about the y-axis.
Find the exact volume of the solid of revolution formed.

1.15 — Question 33.

The diagram below shows the curve

$$x = k\sqrt{y} - 0.05y^2$$

for $0 \leq y \leq 16$ and part of the line $x = 3$



The point $P(3.2, 16)$ lies on the curve.

a) Determine the value of k

When rotated 2π radians about the y-axis it forms a solid, which may be used to model the volume of a large vase. The vase has a height of 18 cm, where 1 unit represents 1 cm.

b) Find the total volume of the solid created giving your answer to 3 significant figures.

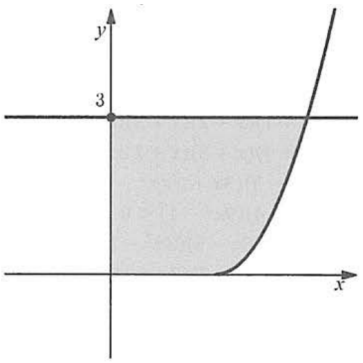
c) State a limitation of using this model

i) for the vase.

ii) to find the volume of the vase.

1.15 — Question 34.

The diagram below shows the region enclosed by part of the curve $y = x^2 - 4x + 4$ for $x \geq 2$ and the x-axis and $y = 3$



The shaded region is rotated 2π radians about the y-axis.
Show that the exact volume of the solid generated is $\left(\frac{33}{2} + 8\sqrt{3}\right)\pi$ units^3.

Mix

Prerequisite knowledge for Q35: Conics

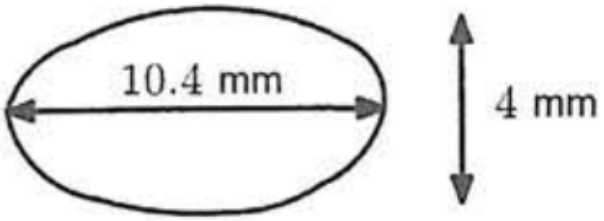
1.15 — Question 35. An ellipse is defined by the equation

$$\frac{x^2}{4} + \frac{y^2}{100} = 1$$

a) Sketch the curve, labelling the points at which it crosses the axes.
b) Find the volume formed by rotating the curve about the x-axis.
c) Find the volume formed by rotating the curve about the y-axis.

1.15 — Question 36.

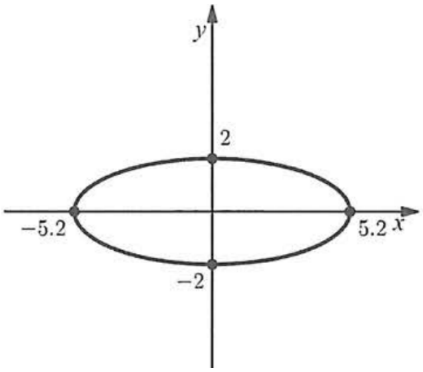
Rebecca wants to find the volume of a small round and slightly flattened chocolate with a hard colourful shell. A diagram of the chocolate from its side is shown below:



It has a length of 10.4 mm and a depth of 4 mm. From the top the chocolate looks circular. Rebecca sketches the curve

$$25x^2 + 169y^2 = 676$$

which she thinks will best model the side of the chocolate, where 1 unit represents 1 mm:



a) Find the volume formed by rotating the curve about the x-axis giving your answer to 3 significant figures.
b) Find the volume formed by rotating the curve about the y-axis giving your answer to 3 significant figures.
c) Which of your answers to part a) and part b) will most closely model the volume of the chocolate?